

Activity Risk Assessment — HTC Vive Motion Tracking System

Project: RoboCup Soccer — Booster K1 (HSL 2026) — External Pose Estimation **ARA Reference:** AILL-Booster-K1-PAR-Development **Equipment:** HTC Vive Base Stations (4× v1.0, 2× v2.0), HTC Vive Trackers (2× v1.0, seeking v2.0), HTC Vive Dongle **Source Documents:**

- docs/equipment/htc_vive_base_station_user_guide.txt
- docs/equipment/htc_vive_tracker_v1.txt
- docs/equipment/htc_vive_tracker_user_manual.txt
- docs/equipment/extracted_activities_for_risk_assessment.md
- docs/ara/.minutes_7_5_26.md

System Architecture:

- The **dongle** connects to the PC via USB cable (dongle → dongle cradle → USB cable → PC).
- The **trackers** communicate wirelessly with the **dongle** (not directly with the PC).
- The **tracker** is mounted on the Booster K1 robot but is **not connected to the robot's control stack**. Tracking data (pose/position) is transmitted wirelessly to the dongle, then via USB to the PC, where it is recorded in ROS2 for external pose estimation analysis.
- The robot is operated independently via its own control system (joystick/teleoperation/autonomous). The tracking system is a passive data-recording layer only.

Risk Score Matrix Reference: Refer to the risk score matrix in the Booster P&ERA: AILL-Booster-K1

Activity 1: Setting up HTC Vive Base Stations on Tripods

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
List the steps required to perform the activity or task in the sequence they are carried out.	List the hazards and risks that could cause injury when the activity or task is performed	Detail the controls currently in place that will reduce the risk. If none exist, please note this	Consequence	Likelihood	Risk Rating	List the control measures required to eliminate or further minimise the risk of injury/incident arising

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
1. Transport base stations and tripods to the VXLab or testing area.	Manual Handling: Ergonomic from repeated bending/lifting/carrying equipment.	Comply with the Booster P&ERA: AILL-Booster-K1-PAR-Development.	2	C	Medium	Prefer using equipment trolley to transport multiple tripods and base stations.
2. Extend tripod legs and position tripods at designated locations around the tracking area.	Physical Injury: Tripod legs creating tripping hazard for operators and bystanders.	None specific.	2	C	Medium	Mark tripod leg positions with visible tape or cones on the floor. Ensure tripod legs are not placed in walkways or thoroughfares. Retract tripod legs when not in active use.
3. Mount base stations onto tripods using the 1/4" mounting thread and clamping ring.	Physical Injury: Dropping the base station during mounting; pinch hazard from clamping ring. Equipment Damage: Base station falling and shattering if dropped.	Follow mounting instructions in the base station user guide.	2	C	Medium	Ensure two-person operation: one holds the tripod steady, the other mounts the base station. Mount base stations at or below eye level where practical.
4. Adjust base station angle (loosening clamping ring while holding the station).	Physical Injury: Dropping the base station during adjustment; pinch hazard from clamping ring. Equipment Damage: Base station falling and shattering.	Follow angle adjustment instructions in the base station user guide.	2	C	Medium	Ensure the clamping ring is fully tightened before releasing grip on the base station. Use two hands

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
						when adjusting — one supporting the station, one operating the ring.
5. Connect power cables and plug adapters into power outlets.	Electrical: Electric shock; fire risk from damaged cables. Trip Hazard: Power cables across floor.	Use only supplied power cables and adapters. Comply with the Booster P&ERA: AILL-Booster-K1-PAR-Development.	3	C	Medium	Route power cables along tripod legs and use cable covers or tape to secure cables to the floor. Position tripods near power outlets to minimise cable run length. Inspect cables for damage before each use.
6. Power on base stations and verify operational status (green LED).	Electrical: Electric shock from damaged units.	Inspect base station front panel for cracks or damage before powering on.	3	D	Low	—

Activity 2: Setting up HTC Vive Tracker and Dongle

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
List the steps required to perform	List the hazards and risks that could cause injury when the	Detail the controls currently in place that	Consequence	Likelihood	Risk Rating	List the control measures required to eliminate or

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
the activity or task in the sequence they are carried out.	activity or task is performed	will reduce the risk. If none exist, please note this				minimise the risk of injury/incident arising
1. Connect dongle to dongle cradle via USB cable, and connect the other end of the USB cable to the computer.	Trip Hazard: USB cable routed across workspace.	None specific.	2	C	Medium	Route USB cable along desk edges or use cable clips. Keep cable length as short as practical. Place dongle cradle in a location where the cable does not cross walkways.
2. Place dongle at least 45 cm (18 in) from the computer.	RF Interference: Tracking dropout or degraded performance if dongle is too close to computer.	Follow dongle placement instructions in the tracker user manual (≥ 45 cm from computer).	2	C	Medium	Mark the dongle cable at 45 cm from the dongle to indicate minimum distance. Ensure the dongle is placed on a stable surface where it will not be moved.
3. Attach tracker to the Booster K1 robot using the 1/4" screw mounting mechanism and stabilizing pin.	Physical Injury: Pinch hazard from docking mechanism. Equipment Damage: Dropping the tracker; over-tightening causing damage; stripping threads.	Follow tracker mounting instructions in the tracker user manual and v1 technical reference.	2	C	Medium	Ensure the tracker is securely fastened before releasing grip. Do not over-tighten the 1/4" screw — tighten only until snug. Verify the stabilizing pin is fully engaged in its recess.
4. Verify tracker placement	Equipment Damage: Accessory body	Follow placement guidelines in	2	C	Medium	Visually inspect that the tracker's sensor dome has

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does not obstruct robot sensors or movement.	obstructing robot sensors or limiting range of motion. Tracking Degradation: Accessory extending beyond recommended placing cone (270° FOV) blocking sensor view.	the tracker v1 technical reference.				a clear line of sight to the base stations. Verify the tracker is within the 270° FOV cone. Test robot range of motion after mounting to ensure no obstruction.
5. Power on tracker (briefly press Power button).	Operational: Tracker may automatically turn off if not paired within 30 seconds.	None specific.	1	D	Low	—
6. Pair tracker with dongle via SteamVR app (press Power button for ~2 seconds).	RF Interference: Potential RF interference in lab environment.	Follow pairing instructions in the tracker user manual.	1	D	Low	—
7. Verify connection status (green LED = normal, solid green in SteamVR).	Operational: Misinterpreting LED status could lead to operating with low battery or in pairing mode.	Refer to LED status indicator table in tracker user manual.	1	D	Low	—

Activity 3: Charging HTC Vive Trackers

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1. Connect USB cable to tracker micro-USB port.	Electrical: Electric shock if cable or port is damaged.	Use only the supplied USB cable.	2	D	Low	—
2. Connect USB cable to power adapter (5V $\pm$ 5%, 1000 mA) and plug into power outlet.	Electrical: Overheating if using incorrect charger; fire risk.	Use only the power adapter that came with VIVE controllers (5V $\pm$ 5%, 1000 mA). Comply with the Booster P&ERA: AILL-Booster-K1-PAR-Development.	3	C	Medium	Do not leave charging trackers unattended for extended periods. Charge on a non-flammable surface away from combustible materials. Inspect charger and cable for damage before each use.
3. Alternatively, connect USB cable to PC USB port (500 mA).	Electrical: USB port overload if multiple devices connected.	Follow charging instructions in the tracker user manual.	2	D	Low	—
4. Monitor charging status (orange LED while charging;	Operational: Misinterpreting LED status leading to overcharging or operating with	Refer to LED status indicator table in tracker user manual.	1	D	Low	—

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
green/white when fully charged).	insufficient charge.					
5. Disconnect USB cable once fully charged.	Trip Hazard: USB cable left across workspace after disconnection.	None specific.	2	C	Medium	Coil and store USB cable immediately after disconnection.

Activity 4: Operating the HTC Vive Motion Tracking System (Vive ROS2 Integration)

Note: The tracker is mounted on the robot but is **not connected to the robot's control system**. Tracking data (pose/position) is transmitted wirelessly to the dongle, then via USB to the PC, where it is recorded in ROS2 for external pose estimation analysis. The robot is operated independently. Therefore, tracking issues (data dropout, signal loss, etc.) affect only data recording, not robot behaviour.

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1. Ensure the tracking area (play area) is entirely clear of obstacles, objects, and other individuals.	Physical Injury: Collision with obstacles or other individuals in the tracking space while the robot is in motion.	Clear the play area before operation as per tracker user manual. Comply with the Booster P&ERA: AILL-Booster-K1-	3	C	Medium	Conduct a visual sweep of the tracking area before each session. Erect physical barriers or warning tape around the

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		PAR-Development.				perimeter of the tracking area. Only permitted personnel may be within the tracking area during operation.
2. Power on the Booster K1 robot (with tracker attached) according to the robot ARA.	As defined in the Booster K1 ARA (AIII-Booster-K1-PAR-Development).	Comply with the Booster P&ERA: AIII-Booster-K1-PAR-Development.	—	—	—	—
3. Launch the Vive ROS2 integration software on the PC to begin receiving and recording tracking data.	Software: ROS2 node misconfiguration causing failure to record tracking data.	None specific.	1	C	Low	Verify that all base stations are powered on and trackers are detected (green LED) before launching ROS2. Confirm ROS2 topics are publishing data before proceeding with recording.
4. Verify tracking data is being received correctly (check ROS2 topics for pose data).	Data Loss: Incorrect or noisy tracking data leading to unusable recordings.	None specific.	1	C	Low	Monitor ROS2 tracking topics for data validity. Define acceptable bounds for tracking data (e.g., position, orientation). If tracking data is out of bounds, troubleshoot

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
						before continuing.
5. Operate the robot (teleoperation or autonomous) according to the robot ARA while the tracking system records pose data.	As defined in the Booster K1 ARA (AIII-Booster-K1-PAR-Development). The tracking system is passive and does not affect robot control.	Comply with the Booster P&ERA: AIII-Booster-K1-PAR-Development. Hold and have ready access to the joystick for E-Stop. Always monitor the K1.	—	—	—	—
6. Monitor for tracking degradation during operation (bright light, reflective surfaces).	Data Loss: Reduced tracking accuracy due to bright light or reflective surfaces, resulting in poor-quality recordings.	Be aware of environmental factors that affect tracking performance.	1	C	Low	Position base stations away from direct bright light sources. Avoid reflective surfaces in the tracking area.
7. Monitor tracker battery level during operation.	Data Loss: Tracker automatically turning off due to low battery, pairing timeout (>30 s idle), or inactivity (>5 min no movement), causing loss of tracking data.	Monitor tracker LED status (blinking red = low battery).	1	C	Low	Check tracker battery level before each session. Keep spare charged trackers available for swap. If tracker turns off unexpectedly, the robot continues operating normally — only tracking data recording is affected.

Activity 5: Updating Firmware (Base Stations and Trackers)

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1. Connect base station to computer via micro-USB cable (for base station firmware update).	Trip Hazard: USB cable across workspace.	None specific.	2	C	Medium	Keep USB cable routed along desk edges. Ensure the cable is not across walkways.
2. Connect tracker to computer via supplied USB cable (for tracker firmware update).	Trip Hazard: USB cable across workspace.	None specific.	2	C	Medium	Keep USB cable routed along desk edges. Ensure the cable is not across walkways.
3. Initiate firmware update via SteamVR app (tracker) or Bluetooth/USB (base station).	Equipment Damage: Bricking the device if power or USB is disconnected during update.	Do not unplug the USB cable or disconnect power any time before the firmware update is complete. Follow firmware update instructions in the respective user guides.	3	C	Medium	Ensure the computer has stable power (plugged in, not on battery). Do not put the computer to sleep during the update. Verify the correct firmware file before proceeding. Allow the update to complete

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
						without interruption.
4. Verify firmware update completed successfully (check device status).	Equipment Damage: Incorrect firmware file causing device malfunction.	Follow verification steps in the user guides.	2	D	Low	—

Activity 6: Packing Up and Storing HTC Vive Equipment

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
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1. Power off base stations and disconnect power cables.	Electrical: Electric shock if cables are pulled forcefully.	Power off before disconnecting.	2	D	Low	—
2. Unmount base stations from tripods.	Physical Injury: Dropping the base station during unmounting. Equipment Damage:	Use two-person operation: one holds the tripod steady, the other	2	C	Medium	Ensure the base station is securely held before loosening the clamping ring.

Activity or Task	Hazards & Risks	Current Risk Controls	Current Risk Rating (with existing controls)	Additional Risk Control Measures	Residual Risk Rating (after additional controls)	Responsibility
	Base station falling and shattering.	unmounts the base station.				Place the base station in its protective case immediately after unmounting.
3. Collapse tripod legs and pack tripods.	Physical Injury: Pinch hazard from collapsing tripod legs.	None specific.	2	C	Medium	Collapse tripod legs one section at a time, keeping fingers clear of locking mechanisms.
4. Remove tracker from Booster K1 robot (if applicable).	Physical Injury: Pinch hazard from docking mechanism. Equipment Damage: Dropping the tracker.	Follow tracker removal instructions.	2	C	Medium	Place the tracker in its protective case immediately after removal.
5. Disconnect and store dongle, cables, and accessories.	Trip Hazard: USB cable left across workspace after disconnection.	None specific.	2	C	Medium	Coil and store all cables immediately after disconnection. Store dongle in its cradle or protective case.
6. Transport equipment to storage location.	Manual Handling: Ergonomic from repeated bending/lifting/carrying equipment.	Comply with the Booster P&ERA: AILL-Booster-K1-PAR-Development. Comply with RMIT manual handling policies.	2	C	Medium	Use equipment trolley for multiple items. Do not carry more than one tripod at a time. Bend with knees and keep a straight back when lifting.

Risk Score Matrix Reference

Refer to the risk score matrix in the Booster P&ERA: AILL-Booster-K1-PAR-Development.

Consequence	Description
1	Minor — No injury or negligible damage
2	Moderate — Medical attention required, reversible damage
3	Major — Serious injury, significant equipment damage
4	Severe — Fatality, permanent disability, catastrophic damage

Likelihood	Description
A	Almost Certain — Expected to occur in most circumstances
B	Likely — Will probably occur in most circumstances
C	Possible — Might occur at some time
D	Unlikely — Could occur but not expected
E	Rare — May occur only in exceptional circumstances

Risk Rating	Action Required
Low	Acceptable — No further controls required, but monitor for changes
Medium	Tolerable — Additional controls should be implemented where reasonably practicable
High	Unacceptable — Activity must not proceed until controls are implemented to reduce risk
Extreme	Intolerable — Activity must not proceed